

Supplementary Table 4: Sex differences in cross-trait genetic correlations

Pair	rg(se) males	rg(se) females	P diff	P diff adj
HDL—LDL	0.1636 (0.0238)	-0.1122 (0.0238)	2.524336e-16	1.128378e-13
LDL—Waist circumference	-0.1423 (0.0311)	0.1842 (0.028)	6.086319e-15	1.360292e-12
Educational attainment—Risk-taking behaviour	-0.1432 (0.0341)	0.187 (0.0335)	4.928616e-12	7.343638e-10
Body Mass Index—LDL	-0.0743 (0.0246)	0.1457 (0.0232)	7.710497e-11	8.61648e-09
LDL—nonHDL	0.8852 (0.0113)	0.9604 (0.0036)	2.284903e-10	2.042703e-08
LDL—Waist-hip ratio	-0.0068 (0.0356)	0.2813 (0.0338)	4.388372e-09	2.802289e-07
LDL—logTG	0.126 (0.032)	0.362 (0.0242)	4.04589e-09	2.802289e-07
Waist circumference (adjBMI)—Waist-hip ratio	0.3631 (0.0392)	0.6302 (0.0284)	3.431891e-08	1.917569e-06
LDL—Weight	-0.1565 (0.0312)	0.0886 (0.0319)	3.953715e-08	1.963679e-06
TC—Waist circumference	-0.1672 (0.0309)	0.0526 (0.0268)	7.713816e-08	3.448076e-06
nonHDL—Waist circumference	0.0197 (0.036)	0.2637 (0.0295)	1.584455e-07	6.438648e-06
LDL—Type 2 diabetes	-0.0144 (0.0305)	0.1879 (0.0262)	4.870976e-07	1.814439e-05
Hip circumference (adjBMI)—Waist circumference (adjBMI)	0.6185 (0.03)	0.3465 (0.0465)	8.865402e-07	3.048334e-05
Body Mass Index—Waist-hip ratio	0.6636 (0.028)	0.4435 (0.0359)	1.335649e-06	4.264536e-05
Body Mass Index—Weight	0.7813 (0.0186)	0.8867 (0.0134)	4.270643e-06	1.272652e-04
Waist-hip ratio—Waist-hip ratio (adjBMI)	0.7266 (0.025)	0.8568 (0.0137)	4.943196e-06	1.381005e-04
Hip circumference—LDL	-0.1518 (0.0356)	0.0489 (0.0274)	7.910357e-06	2.079959e-04
LDL—Waist circumference (adjBMI)	-0.0913 (0.0331)	0.1226 (0.0349)	8.708763e-06	2.162676e-04
TC—Weight	-0.201 (0.0314)	-0.0114 (0.0298)	1.187923e-05	2.794746e-04
Age at first birth—Risk-taking behaviour	-0.3384 (0.0542)	-0.0351 (0.0463)	2.091942e-05	4.67549e-04
Waist circumference (adjBMI)—Waist-hip ratio (adjBMI)	0.5318 (0.0365)	0.7171 (0.0243)	2.380296e-05	5.06663e-04
nonHDL—Waist circumference (adjBMI)	-0.0365 (0.033)	0.1634 (0.0346)	2.904842e-05	5.90211e-04
HDL—nonHDL	-0.1165 (0.0291)	-0.2742 (0.0247)	3.602685e-05	7.00174e-04
Body Mass Index—nonHDL	0.0634 (0.0276)	0.2174 (0.0261)	5.033248e-05	9.374425e-04
Body Mass Index—TC	-0.0972 (0.0239)	0.0314 (0.0212)	5.68931e-05	1.017249e-03
logTG—Waist circumference (adjBMI)	0.1023 (0.0361)	0.2879 (0.0296)	7.017556e-05	1.20648e-03
Height—Weight	0.584 (0.0289)	0.4191 (0.0308)	9.450444e-05	1.564574e-03
Risk-taking behaviour—Schizophrenia	0.1606 (0.0323)	0.355 (0.0389)	1.206525e-04	1.907459e-03
nonHDL—Weight	-0.0452 (0.034)	0.1391 (0.0339)	1.237502e-04	1.907459e-03
Insomnia—Schizophrenia	0.1434 (0.0356)	-0.0536 (0.0374)	1.360288e-04	2.026829e-03
nonHDL—Waist-hip ratio	0.1794 (0.0435)	0.3862 (0.0334)	1.627713e-04	2.347057e-03
LDL—Waist-hip ratio (> 50 yrs, adjBMI)	0.0123 (0.0467)	0.2464 (0.0417)	1.846582e-04	2.579444e-03
Hip circumference—TC	-0.2025 (0.0347)	-0.0453 (0.0241)	1.985302e-04	2.689182e-03
Hip circumference—Waist circumference (adjBMI)	0.3289 (0.0434)	0.1208 (0.0355)	2.060731e-04	2.709256e-03
HDL—Waist circumference (adjBMI)	-0.1274 (0.031)	-0.2786 (0.0283)	3.155901e-04	4.030536e-03
Free thyroxine (FT4)—Waist-hip ratio (adjBMI)	0.1877 (0.0767)	-0.1967 (0.0748)	3.332332e-04	4.137645e-03
HDL—Height SDS	-0.1902 (0.0502)	0.0363 (0.0386)	3.478112e-04	4.201935e-03
Waist circumference (adjBMI)—Waist-hip ratio (> 50 yrs, adjBMI)	0.5401 (0.0436)	0.7287 (0.0312)	4.351808e-04	5.119101e-03
Body Mass Index—Hip circumference	0.7943 (0.0204)	0.8768 (0.0117)	4.513223e-04	5.172848e-03
ADHD—Schizophrenia	0.1855 (0.0413)	-0.1203 (0.0784)	5.585882e-04	6.242223e-03
Fasting insulin—LDL	-0.1308 (0.0574)	0.1671 (0.0661)	6.668535e-04	7.27033e-03
Educational attainment—Insomnia	-0.1996 (0.0426)	-0.3747 (0.0299)	7.672652e-04	8.165894e-03

Height—Hip circumference	0.4591 (0.0375)	0.3001 (0.0291)	8.088824e-04	8.408615e-03
Type 2 diabetes—Waist circumference (adjBMI)	0.0856 (0.0425)	0.2774 (0.0391)	8.962699e-04	9.105287e-03
TC—Waist-hip ratio	-0.0067 (0.0366)	0.1526 (0.0318)	1.017839e-03	9.890737e-03
Waist circumference (adjBMI)—Waist-hip ratio (<= 50 yrs, adjBMI)	0.5818 (0.0616)	0.8398 (0.0486)	1.008469e-03	9.890737e-03
LDL—Waist-hip ratio (adjBMI)	0.0547 (0.0407)	0.2278 (0.0352)	1.295989e-03	1.232568e-02
Schizophrenia—Snoring	-0.2016 (0.0312)	-0.0598 (0.032)	1.509892e-03	1.406087e-02
Educational attainment—Multisite chronic pain	-0.5398 (0.029)	-0.4193 (0.0247)	1.559935e-03	1.423043e-02
Number of children ever born—Risk-taking behaviour	0.4164 (0.0719)	0.1133 (0.0641)	1.651505e-03	1.476446e-02
TC—Waist circumference (adjBMI)	-0.0837 (0.0298)	0.0583 (0.0342)	1.745702e-03	1.530056e-02
Hip circumference—nonHDL	-0.0676 (0.0397)	0.0907 (0.0318)	1.85755e-03	1.596778e-02
Cardiovascular disease—Waist circumference (adjBMI)	0.0028 (0.0367)	0.1863 (0.0466)	1.977619e-03	1.667916e-02
Height SDS—Weight	0.7631 (0.0868)	0.4633 (0.0496)	2.710098e-03	2.243359e-02
Grip strength measurement/ Muscle weakness—Left 5m area	-0.1617 (0.1184)	0.2609 (0.0795)	3.044198e-03	2.474103e-02
Height—Waist-hip ratio (adjBMI)	-0.1432 (0.0497)	0.0364 (0.0379)	4.059407e-03	3.143235e-02
logTG—nonHDL	0.4823 (0.0256)	0.5755 (0.0199)	4.04873e-03	3.143235e-02
nonHDL—Type 2 diabetes	0.1914 (0.0303)	0.309 (0.0276)	4.114148e-03	3.143235e-02
nonHDL—Waist-hip ratio (> 50 yrs, adjBMI)	0.1353 (0.0496)	0.3176 (0.0398)	4.148789e-03	3.143235e-02
Height—Hip circumference (adjBMI)	0.8468 (0.0251)	0.751 (0.0224)	4.404351e-03	3.281241e-02
ADHD—Age at first birth	-0.4584 (0.0637)	-0.7776 (0.0925)	4.481914e-03	3.284287e-02
Schizophrenia—Waist-hip ratio (> 50 yrs, adjBMI)	0.0965 (0.039)	-0.0759 (0.0479)	5.253836e-03	3.787846e-02
HDL—Waist-hip ratio (adjBMI)	-0.2234 (0.0424)	-0.3695 (0.0309)	5.35745e-03	3.801238e-02
Educational attainment—Waist-hip ratio (> 50 yrs, adjBMI)	-0.2735 (0.0443)	-0.1095 (0.0397)	5.834276e-03	3.898618e-02
Height—Waist-hip ratio	-0.1691 (0.0433)	-0.0138 (0.0358)	5.706586e-03	3.898618e-02
Height—Waist-hip ratio (> 50 yrs, adjBMI)	-0.0994 (0.0583)	0.0958 (0.0402)	5.843566e-03	3.898618e-02
logTG—Waist-hip ratio (adjBMI)	0.3145 (0.038)	0.451 (0.0316)	5.746375e-03	3.898618e-02
HDL—Snoring	-0.1587 (0.0262)	-0.2534 (0.0226)	6.2012e-03	4.076377e-02
Height SDS—Hip circumference	0.6941 (0.0905)	0.4018 (0.0589)	6.789255e-03	4.398256e-02
logTG—Waist-hip ratio	0.4629 (0.0374)	0.5933 (0.0305)	6.891529e-03	4.400733e-02
Fasting insulin—TC	-0.1319 (0.0548)	0.0918 (0.0634)	7.598222e-03	4.783669e-02
logTG—Waist-hip ratio (<= 50 yrs, adjBMI)	0.289 (0.0476)	0.4695 (0.0507)	9.445044e-03	5.863798e-02
Snoring—Weight	0.1709 (0.0378)	0.3014 (0.0339)	1.016434e-02	6.223919e-02
Fasting insulin—nonHDL	0.0542 (0.0663)	0.3008 (0.0698)	1.042e-02	6.294241e-02
Body Mass Index—Insomnia	0.0391 (0.043)	0.1731 (0.0313)	1.17526e-02	6.948323e-02
Pubertal growth—Weight	0.0656 (0.0776)	-0.2251 (0.0855)	1.18137e-02	6.948323e-02
Cardiovascular disease—Left 5m area	-0.0752 (0.0748)	0.2096 (0.0857)	1.229053e-02	7.134893e-02
Educational attainment—Schizophrenia	-0.0071 (0.0286)	0.0985 (0.0312)	1.259631e-02	7.218653e-02
Age at first birth—Carotid intima-media thickness	-0.2637 (0.0806)	-0.0123 (0.0625)	1.370628e-02	7.620274e-02
Height SDS—Waist circumference	0.5717 (0.0918)	0.3115 (0.0524)	1.383081e-02	7.620274e-02
Insomnia—Word reading	-0.0197 (0.0852)	-0.3328 (0.0948)	1.403155e-02	7.620274e-02
Number of children ever born—Snoring	-0.0356 (0.0646)	0.1593 (0.0462)	1.412654e-02	7.620274e-02
Schizophrenia—Type 2 diabetes	-0.0432 (0.0268)	-0.1475 (0.033)	1.41495e-02	7.620274e-02
Grip strength measurement/ Muscle weakness—Waist-hip ratio (adjBMI)	0.2622 (0.0791)	0.0365 (0.0497)	1.569095e-02	8.349828e-02

HDL—Waist-hip ratio (≤ 50 yrs, adjBMI)	-0.2138 (0.0452)	-0.3671 (0.0456)	1.695739e-02	8.917591e-02
HDL—TC	0.2388 (0.0243)	0.1579 (0.024)	1.785127e-02	9.278507e-02
HDL—Waist-hip ratio (> 50 yrs, adjBMI)	-0.2251 (0.0427)	-0.3518 (0.0334)	1.943102e-02	9.983523e-02
Hip circumference (adjBMI)—Risk-taking behaviour	-0.16 (0.0388)	-0.0337 (0.038)	2.004031e-02	1.017957e-01
Type 2 diabetes—Weight	0.3839 (0.0355)	0.503 (0.0377)	2.145068e-02	1.077355e-01
Educational attainment—LDL	-0.0813 (0.0237)	-0.1516 (0.02)	2.339461e-02	1.154866e-01
Fasting insulin—Free thyroxine (FT4)	-0.253 (0.0983)	0.1436 (0.1449)	2.351069e-02	1.154866e-01
Body Mass Index—Left Caudate	0.1328 (0.0456)	-0.0034 (0.0402)	2.5058e-02	1.180377e-01
Height SDS—Waist circumference (adjBMI)	0.5973 (0.084)	0.358 (0.066)	2.50863e-02	1.180377e-01
Hip circumference—Pubertal growth	0.0253 (0.0944)	-0.2397 (0.0708)	2.471921e-02	1.180377e-01
TC—Waist-hip ratio (> 50 yrs, adjBMI)	0.0503 (0.0456)	0.1872 (0.0404)	2.463194e-02	1.180377e-01
Waist circumference (adjBMI)—Weight	0.3461 (0.0402)	0.2247 (0.037)	2.628346e-02	1.198848e-01
logTG—Waist circumference	0.3483 (0.0334)	0.4453 (0.028)	2.604163e-02	1.198848e-01
logTG—Waist-hip ratio (> 50 yrs, adjBMI)	0.3044 (0.0452)	0.4315 (0.0348)	2.587496e-02	1.198848e-01
Body Mass Index—Type 2 diabetes	0.4581 (0.0319)	0.5541 (0.0293)	2.666583e-02	1.204003e-01
TC—Type 2 diabetes	0.0128 (0.0264)	0.0941 (0.0258)	2.763343e-02	1.235214e-01
Fasting insulin—Waist circumference (adjBMI)	0.0687 (0.0981)	0.367 (0.0939)	2.804463e-02	1.241183e-01
Insomnia—Left Caudate	0.0217 (0.0619)	0.1955 (0.0496)	2.844414e-02	1.246522e-01
Educational attainment—Waist-hip ratio	-0.3887 (0.0362)	-0.2822 (0.0327)	2.902396e-02	1.255817e-01
Fasting glucose—LDL	-0.169 (0.0561)	0.009 (0.0593)	2.921812e-02	1.255817e-01
Weight—Word reading	0.0336 (0.0763)	-0.2046 (0.08)	3.118947e-02	1.32778e-01
ADHD—Insomnia	0.2676 (0.0582)	0.4836 (0.0825)	3.240316e-02	1.337902e-01
ADHD—Type 2 diabetes	0.3697 (0.0412)	0.2086 (0.0633)	3.292377e-02	1.337902e-01
Body Mass Index—Waist-hip ratio (≤ 50 yrs, adjBMI)	-0.0175 (0.0513)	-0.1707 (0.0499)	3.229987e-02	1.337902e-01
Insomnia—logTG	0.1342 (0.0302)	0.2209 (0.0271)	3.262189e-02	1.337902e-01
Waist-hip ratio (≤ 50 yrs, adjBMI)—Waist-hip ratio (> 50 yrs, adjBMI)	0.7853 (0.102)	1.0633 (0.0809)	3.272998e-02	1.337902e-01
nonHDL—Waist-hip ratio (adjBMI)	0.1765 (0.0472)	0.3017 (0.0351)	3.329482e-02	1.340791e-01
Grip strength measurement/ Muscle weakness—Waist-hip ratio (> 50 yrs, adjBMI)	0.2158 (0.0898)	-0.014 (0.0607)	3.399683e-02	1.356838e-01
Multisite chronic pain—Risk-taking behaviour	0.2851 (0.0446)	0.1618 (0.0376)	3.454345e-02	1.366453e-01
Age at first birth—HDL	0.2498 (0.0312)	0.3367 (0.0277)	3.726661e-02	1.461244e-01
Cardiovascular disease—LDL	0.0918 (0.0354)	0.1982 (0.037)	3.772413e-02	1.46632e-01
Fasting insulin—Waist-hip ratio (adjBMI)	0.0857 (0.098)	0.3841 (0.1059)	3.863092e-02	1.488622e-01
Cardiovascular disease—Hip circumference (adjBMI)	-0.2352 (0.0398)	-0.1095 (0.0462)	3.926929e-02	1.500288e-01
Body Mass Index—Risk-taking behaviour	0.1353 (0.0389)	0.0308 (0.0329)	4.025312e-02	1.524843e-01
Educational attainment—logTG	-0.2776 (0.0229)	-0.3438 (0.0229)	4.094117e-02	1.525059e-01
Hip circumference—Hip circumference (adjBMI)	0.5998 (0.0315)	0.5176 (0.0249)	4.064123e-02	1.525059e-01
Fasting insulin—Waist-hip ratio (> 50 yrs, adjBMI)	0.1622 (0.1133)	0.4965 (0.1184)	4.135501e-02	1.527743e-01
Age at first birth—Insomnia	-0.2036 (0.0632)	-0.3525 (0.0378)	4.31809e-02	1.58212e-01
Educational attainment—Waist-hip ratio (adjBMI)	-0.2582 (0.037)	-0.1572 (0.0337)	4.358002e-02	1.583762e-01
HDL—Insomnia	-0.1138 (0.029)	-0.1918 (0.0257)	4.41197e-02	1.590444e-01
Grip strength measurement/ Muscle weakness—Waist-hip ratio	0.2592 (0.0719)	0.0884 (0.0458)	4.511699e-02	1.600733e-01
Grip strength measurement/ Muscle weakness—Weight	-0.2306 (0.0602)	-0.0803 (0.0448)	4.518611e-02	1.600733e-01

Hip circumference (adjBMI)—Weight	0.5127 (0.0412)	0.4086 (0.0318)	4.547944e-02	1.600733e-01
Educational attainment—Weight	-0.1056 (0.0361)	-0.1988 (0.0296)	4.588897e-02	1.602529e-01
LDL—Schizophrenia	0.0662 (0.0228)	-0.0031 (0.0263)	4.648346e-02	1.610706e-01
Educational attainment—Snoring	-0.1327 (0.034)	-0.2223 (0.0299)	4.78241e-02	1.644413e-01
Type 2 diabetes—Waist circumference	0.4995 (0.0419)	0.601 (0.0298)	4.8373e-02	1.65059e-01
Multisite chronic pain—Waist-hip ratio (> 50 yrs, adjBMI)	0.1694 (0.0555)	0.0319 (0.0424)	4.898661e-02	1.658865e-01